

Picture Mapping

1. Feature detection & extraction: SuperPoint
2. Stitching - SuperGlue
3. Spatial representation creation - Structure from Motion

Background (On Device)

VIO for location tracking. Keeps the dot live while the process below sends in updates.

On Edge

Offline

Superpoint | SemaGlue Semantic Descriptor Creator

SegaGlue Fusion Block

FASTMAP Point Cloud Creation

SuperVLAD embedding archive

Paired semantic descriptors

Online

Incorporate a privacy blurring mechanism

This could be that semantic negative mask.

Pose/Localization

Find a limiting mechanism to account for known position.

Odometry should actually keep most jumps at bay. The initial localization will be difficult however.

Course

FAISS for large dataset compatibility

SuperVLAD (Output is 10-100 images based on scene size)

Text/OCR Filter (TextinPlace inspired)

Fine

Superpoint | SemaGlue Semantic Descriptor Creator

SemaGLUE matching mechanism

Attention mechanism combining glue descriptors with semantic descriptors

Pose

Tracking

PnP + Ransac

Map Refinement (Parallel w/ above)

Loop Closure

Not yet decided

Map Update Mechanism

Route Planning

Figure this out

Guidance through Visual to Auditory Pipeline

Trash on the right in 10 feet

Routing (The agentic kind)

I could not find a meaningful incorporation here. The models that could create substantial improvement are too heavy. Wait actually the VIO vs VPR is kind of this. More parallel.

Cou

Experimental Upgrades:

Borrow text to semantic patch landmark from TEXTSLAM

Introduce a text filter between course and fine (TextinPlace: Indoor VPR | Repetitive Structures)

with Scene Text Spotting and Verification

Incorporate Semantics via SemaGlue-style attention block bridge

Can semantic segmentation be borrowed from a PANSi3R type approach?

Alternative

Incorporate Semantic & Text Embeddings into SuperGlue input. Retrain to perform on this type of input.

OpenGlue (2022)

Shows that SuperGlue-like graph matchers can be retrained with additional descriptor-side information such as scale, orientation, and affine geometry. It raises the idea that extra non-visual channels can be injected into a SuperGlue-style matcher.

Leveraging Semantic Cues from Foundation Vision Models for Enhanced Local Feature Correspondence (2024)

29% improvement in camera localization and comparable accuracy to LightGlue/LoFTR by semantically conditioning existing local descriptors using foundation-model features such as DINOv2.

SemaGlue (2025)

LightGlue-style semantic feature matcher that that injects semantic priors via an attention block that aligns semantic and geometric/local-feature representations.

NO ONE HAD DONE TEXT EMBEDDINGS

OCR Landmarks have been created and incorporated into [loop closures](#)(2015) and [TEXTSLAM](#)(2024).

ORBSLAM

Online oriented

Faster than SuperPoint/SuperGlue

Mobile Native

Sparsier!

Completely different internal components than both SfM and feedforward transformer SLAM

FAST generates the ORB interest points.

BRIEF/ORB generates the descriptors.

Hamming-distance descriptor matching does the low-level matching.

Triangulation generates the sparse 3D map points.

PnP estimates camera pose from 2D–3D matches.

Bundle adjustment refines the poses and map.

Bag-of-words over ORB descriptors supports loop closure.

SfM better for larger maps

match more image pairs

build longer feature tracks

discard bad matches more carefully

run larger bundle adjustment

reprocess/retry weak areas

Location Determination

Does location need to be determined as of anywhere in the hallway or can it be determined as to whether it matches a specific set of points? (corners, segmented straight-aways, etc)

LOOK INTO CONFIDENCE AND UNCERTAINTY ESTIMATIONS

RESEARCH AND ADD IN A MAP UPDATE STEP FOR THE EDGE PORTION

Smart Glasses for accessibility

FIGURE OUT HOW TO INCORPORATE VISUAL LANDMARK MAPPING

Overarching Mission/Purpose

What ties this all together? The goal of mobile ready assistive navigation technology for feature poor locations such as offices, universities, and hospitals.

XMAP focus for mobile ready efficiency. Incorporation of text/OCR for improved FPE mapping/localization. Semantic features for meaningful human navigations and directions.

More questions:

Can odometry be incorporated here to improve the map?

Where splitting can occur:

1. Image storage on the cloud?

2. Live map refinement & loop closure

3. Privacy filtering (Maybe local to not transmit?)

4. Building of VPR/FAISS indexes

5. CONFIDENCE ESTIMATION

a. Activate heavier models when needed

Edge sends corrections, loop closures, semantic updates, or map updates when ready.

Not expected:

A potential pipeline could be just VPR + *light matching*? + PnP

If weak:

edge runs LightGlue / semantic OCR re-ranking / denser matching

If still weak:

edge runs map update, dense alignment, or human-safe fallback guidance

VPR alternative to NetVLAD

CosPlace 2022 - fast query post embedding creation

EigenPlaces 2023 - 50% smaller descriptors, 60% less GPU training

SuperVLAD 2024

TextinPlace-style OCR VPR 2025 - [Explore](#)

Dino, SALAD, AnyLoc

Pros of this approach:

Very scalable

This stack of technologies once proven could map and localize the entire campus.

Robust

Each component technology is published and proven to improve performance.

Novel

This explores multiple new combinations of approaches and technologies.

Unified

Each technology is linked back to increased performance in pursuit of one common goal.

Accessible

Each technology was chosen for mobile ready or common desk/laptop hardware compatible performance.

Service Centric

More than projected SOTA performance in its niche this is dedicated to the goal of improving the human experience of living within our built infrastructure

Best places to add open vocabulary			
Pipeline location	How open vocabulary enters	Value	
VPR / coarse retrieval	Text or language-aligned image embeddings re-rank candidates	Helps retrieve "elevator lobby," "exit hallway," "Room 214"	
Map annotation	Store CLIP/SigLIP embeddings on 3D points, text landmarks, object instances, or regions	Makes the map queryable by language	
Navigation layer	Convert user goals into semantic targets	"Find the nearest elevator," "take me to reception"	
Fine matching	Use language-aligned features inside a SemaGlue-like matcher	Possible, but heavier and less proven	